PYTHAGORAS ▪ SIXFOLD SYMMETRY ▪ EISENSTEIN INTEGERS

The cosine rule at $60°$ and $120°$

From school trigonometry to integer triangles on a hexagonal lattice.

1 One rule, three angles

Let sides a and b meet at angle θ, with side c opposite that angle:

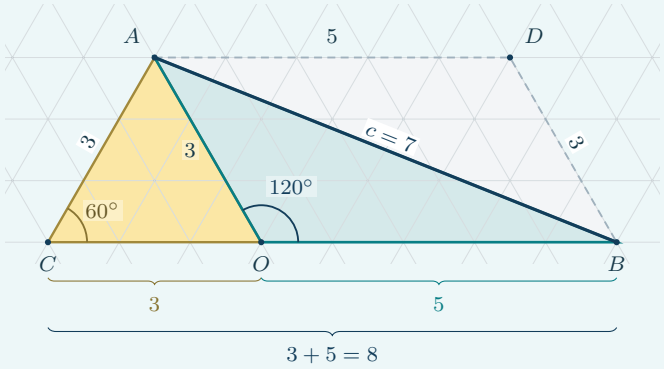
$$\boxed{c^2 = a^2 + b^2 - 2ab\cos\theta}.$$

Included angle	$\cos\theta$	Cosine rule becomes	Example (a,b,c)
$60°$	$\frac{1}{2}$	$c^2 = a^2 - ab + b^2$	$(3,8,7)$
$90°$	0	$c^2 = a^2 + b^2$	$(3,4,5)$
$120°$	$-\frac{1}{2}$	$c^2 = a^2 + ab + b^2$	$(3,5,7)$

The $90°$ case gives Pythagorean triples. The $60°$ and $120°$ cases give **Eisenstein triples**. Here c is opposite the named angle; in $(3,8,7)$, the longest side is $b = 8$.

2 Two triples, one shared side

All marked vertices lie on a unit triangular lattice. The darker triangle OAB has sides $3, 5, 7$.



Gold COA is equilateral. CAB is the $60°$ triangle. The faint copy ADB completes a parallelogram.

$120°$ at O:

$$3^2 + 5^2 + 3\cdot 5 = 49 = 7^2.$$

$60°$ at C:

$$3^2 + 8^2 - 3\cdot 8 = 49 = 7^2.$$

Attaching the equilateral triangle extends the base from 5 to $3+5 = 8$. The same construction works generally:

$$a^2 + (a+b)^2 - a(a+b) = a^2 + ab + b^2.$$

Hence

$$\underbrace{(a,b,c)}_{120°} \longmapsto \underbrace{(a,a+b,c)}_{60°}.$$

Why it tiles. Six $60°$ cells meet at each lattice point. Pair any triangle with its half-turn to make a parallelogram, then repeat by translations. Here the half-turn about the midpoint of AB pairs OAB with ADB, keeping all vertices on the same sixfold lattice.

3 A recipe for more triples

Like Euclid's recipe for Pythagorean triples, this uses two integers. For $m > n > 0$, set

$$a = m^2 - n^2,$$
$$b = 2mn + n^2,$$
$$c = m^2 + mn + n^2.$$

Expanding verifies

$$\boxed{a^2 + ab + b^2 = c^2}.$$

These are $120°$ triples; the construction above supplies their $60°$ companions.

(m,n)	$120°$ triple	$60°$ companion
$(2,1)$	$(3,5,7)$	$(3,8,7)$
$(3,1)$	$(8,7,13)$	$(8,15,13)$

Scaling all 3 sides by same positive integer preserves angle.

4 The Eisenstein connection

The unit step at $120°$ is

$$\omega = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \qquad \omega^3 = 1.$$

Lattice points are **Eisenstein integers**

$$z = m + n\omega, \qquad m, n \in \mathbb{Z}.$$

Their norm is ordinary squared distance:

$$N(z) = |z|^2 = \left(m - \frac{n}{2}\right)^2 + \frac{3n^2}{4} = m^2 - mn + n^2.$$

For a $120°$ triangle, the side c joins the endpoints a and $b\omega$:

$$c^2 = N(a - b\omega) = a^2 + ab + b^2.$$

In the diagram, $A = 3\omega$ and $B = 5$, so

$$AB = |5 - 3\omega| = \sqrt{25 + 15 + 9} = 7.$$

The cross term records the oblique lattice axes.

Reading: OpenStax: law of cosines; S. Chang (2019): integer triples; G. McShane (2024), Section 1.1: Eisenstein integers and their norm.